

Kasi Murali Manoj Vignesh

WEB SECURITY ENGINEER • GENERATIVE AI

🏠 kmmanoj.github.io | 🌐 kmmanoj

Education

Georgia Institute of Technology

M.S. IN COMPUTER SCIENCE

- Computing Systems Specialization

Atlanta, Georgia, US

Aug 2021 - Dec 2023

PES Institute of Technology

B.E. IN COMPUTER SCIENCE AND ENGINEERING

- Gold medalist with a grade score of **9.69** and rank **5**

Bengaluru, Karnataka, India

Aug 2014 - Aug 2018

Industry Experience

Meta

1 Hacker way, Menlo Park,

California, US

SECURITY ENGINEER

01/08/2024 - Present

- **Office Hours:** Partnered with software engineers to assess product and feature architectures, leading threat modeling exercises to identify risks and recommend mitigations. Conducted 50+ follow-up security implementation reviews for high-risk threats to ensure effective control measures.
- **Security Reviews:** Lead security engineer for the products such as AI studio and Business AI agents that prevented 15+ instances of authorization flaws, 10+ instances of prompt injection leaking private information, 5+ instances of integrity issues whose CVSSv3 score is greater than 7.0.
- **Platform security:** Conducted comprehensive security assessments of both products and underlying platforms, identifying 5+ secure design improvements and uncovering 3+ high-impact vulnerabilities.
- **SAST/DAST:** Built 10+ features and improved 25+ features for internal tooling to identify vulnerabilities in web products and platform during development time.
- **Llama firewall:** Contributed to the development of a security framework and 5+ custom scanners; integrated this GenAI firewall into high-risk data processing products to mitigate prompt injection risks across user-facing interfaces.
- **Leadership:** Mentored 2 security engineers. Served as the primary point of contact between the GenAI security team and key product groups including AI Studio, Business AI Agents, and the Data Platform team. Ensured alignment by communicating upcoming features, secure design best practices, and coordinating threat modeling sessions. Delivered 3+ security state presentations to audiences of 20+ stakeholders.

Juniper Networks

1133 Innovation way,

Sunnyvale, California, US

SECURITY ENGINEERING INTERN

05/22/2023 - 08/18/2023

- **Monitoring for Supply chain attack:** Developed a web application that communicates with build tools, and CVE databases (such as NVD, CVEdetails) to monitor CVEs of packages in Juniper software products.
- **Penetration testing:** Contributed to threat modeling, designing test cases and performing VA/PT for Juniper products to find 3 medium or higher vulnerabilities.
- **CTF user training:** Contributed to development of CTF challenges around web, systems and cryptography. The CTF saw 97% registration and the top 5 teams scored close to 90% with average distance between them at 0.8%.

Georgia Institute of Technology

756 West Peachtree Street,

Atlanta, Georgia, US

GRADUATE RESEARCH ASSISTANT

08/22/2022 - 12/08/2023

- **Pirvacy leak due to browser extensions:** Contributed to the research by developing system that taints web elements to track their access and usage by browser extensions.

Georgia Institute of Technology

756 West Peachtree Street,

Atlanta, Georgia, US

GRADUATE TEACHING ASSISTANT

08/22/2022 - 12/08/2023

- **Module lead for SDN:** Presented content and walkthrough project requirements for SDN module of CS 6211: System Design for cloud computing course.
- **Course design:** Improved project evaluations by providing PyTests and better documentation that left room for more deeper interaction during live grading of student's projects.
- **Office hours:** Conducted office hours to nudge students to find answers and answer conceptual questions that students had.
- **Grading:** Live grading of student's work to test student's understand of the subject.

Postman

310, 6th main road,
Indiranagar, Bengaluru,
Karnataka, India

SECURITY ENGINEER

08/11/2021 — 08/12/2022

- **CSPM/CWPM Vendor Assessment:** Assessed multiple CSPM/CWPM vendors against Postman's cloud security requirements. The assessment and the deal improved the visibility of the cloud infrastructure by 3x (based on number of actionable alerts), reduced the cost by about 250% and reduced the maintenance efforts by about 200%.
- **Custom Security Controls:** Developed custom security controls using AWS lambda, Python, Node JS for Postman's AWS environment. This improved the visibility by a further 20%.
- **Threat modelling:** Performed threat modelling at multiple layers of Postman cloud and web application infrastructure. This established a framework to measure security testing coverage and improved the efficiency of VA/PT process.
- **Security Regression testing systems:** Built and improved 7+ automated security regression testing systems to continuously monitor releases of critical services.
- **Vulnerability Assessment and Penetration Testing:** Assessed 50+ releases and 7+ major releases off security vulnerabilities in Identity, Billing and other micro services. Found 5+ security vulnerabilities with CVSS score above 7.
- **Incident Response:** On call support for 100+ hours on web application security and cloud security incidents.
- **Tech Talks:** Presented multiple technology talks on API security - OWASP top 10 security in action themed around a vulnerable demo bank server, Postman as automatic security regression testing platform.

Campus 8, Ecospace, Adarsh
Palm Retreat, Bellandur,
Bengaluru, Karnataka, India

Intuit Inc.

SOFTWARE ENGINEER

08/06/2018 — 08/06/2021

- **Authorization data schema:** Strategized and performed migration of live authorization data to a new schema with zero service downtime.
- **Tech Talks:** Presented multiple technology talks on systems security - Docker container security, AWS resources and infrastructure, threat modelling authorization data access infrastructure.
- **Experimenting FIDO2:** Contributed to experiment of rolling out FIDO2 based multi-factor authentication on the enterprise endpoint systems. Windows hello with Active directory was used for the proof of concept. The security of systems and sign-in user experience of the beta users was observed to have improved.
- **Enterprise Infrastructure Monitoring:** Architected a secure and scalable data pipeline using AWS ECS and AWS lambda and secured with SNMP ACLs and encrypted credentials, from network and video conferencing devices to a monitoring and alerting tool, thereby reducing the MTTD issues by 50%.
- **Chatbot:** Designed, threat modelled and implemented a slack bot using python Django and various AWS services that provides a natural language conversational interface for the automation scripts to the users. Zero security incidents reported till date. It also improved user experience and reduced the learning curve required to understand an automation software.
- **Event Correlation Engine:** Developed a curious and intelligent SIEM like system using python Django in AWS ECS that correlates different network alerts to present an aggregate analysis, thereby further reducing the MTTD by 20%.
- **Infrastructure QA Automation:** Automation of pre-checks and post-checks of the network, video conferencing, and security devices before and after a change. This removed human errors and standardized the test series.
- **Asset Management:** Built a data pipeline using python Django in AWS ECS and AWS lambda for various network and video conferencing orchestration tools to ensure a single source of truth for the asset inventory.
- **Leadership:** Lead a team of 3 engineers to consistently deliver at least 70% of sprint task points. Taught, guided, and advised network engineers on python and AWS that grafted DevOps mindset and enabled the team to write scripts that automate at least 10% of their routine work.
- **Incident response:** On call support for 120+ hours on network and network automation software incidents.

Campus 8, Ecospace, Adarsh
Palm Retreat, Bellandur,
Bengaluru, Karnataka, India

Intuit Inc.

NETWORK AUTOMATION CO-OP

01/08/2018 - 08/03/2018

- **Video Conference troubleshooting:** Developed a web application using React JS and python Django that presents the hop-by-hop network health of each participant of a VC call, which reduced MTTD issues by almost 3 times.

Campus 8, Ecospace, Adarsh
Palm Retreat, Bellandur,
Bengaluru, Karnataka, India

Intuit Inc.

NETWORK AUTOMATION INTERN

05/29/2017 - 07/28/2017

- **VPN troubleshooting:** A CLI utility that gathers VPN client machine data and generates a report that is used by engineers to quickly identify the issue. This reduced the MTTD from 20 minutes to less than a minute

Research & Personal works

Applied Research

<https://kmmanoj.hashnode.dev>

Nov 2018 - Present

BLOGGER ON HASHNODE.COM

- Article on "How complex can Authorization get?"
- ARP Spoofing custom exploit using ScaPy
- Man-in-the-middle via hoax DHCP
- DHCP starvation attack without making any DHCP requests
- Experiment to setup and learn internal network pivoting using Port forwarding
- Burpsuite plugin to capture and query proxy data from a database.
- Learning behavior of exploits and recon using SSH honeypot with cowrie.

Academic Research

- **Arcanum: Detecting and Evaluating the Privacy Risks of Browser Extensions on Web Pages and Web Content (33rd USENIX)**

A dynamic taint tracking system for modern Chrome extensions designed to monitor the flow of user content from web pages [<https://www.usenix.org/conference/usenixsecurity24/presentation/xie-qinge>]

- **Modelling Trust Frameworks for Network-IDS (INCET 2021 - IEEE)**

Mathematically modelling human instinctive trust among systems to communicate security based trust reduction and redemption messages. [<https://doi.org/10.1109/INCET51464.2021.9456381>]

- **Tamper Detection and Correction of ELF Binaries by Remote Parties via Merkle Trees (ICCISoT 2020 - Springer)**

Efficiently identifying and patching malicious software using merkle trees before the first execution by the user. [https://doi.org/10.1007/978-3-030-66763-4_14]

Center for Cloud computing & Big Data, PES Institute of Technology

Bengaluru, Karnataka, India

RESEARCH INTERN

Jun 2016 - Dec 2017

- **Green Cloud:** Energy Efficient datacenter management under availability constraints. To design a VM allocation, migration, and scheduling policy to ensure high availability while optimizing the amount of energy consumed by the data center.
- **Digital Swachh Bharat:** Digital initiative to clean India. To research strategic placement of garbage bins and efficient track for garbage truck based on multiple parameters such as size of truck, location and volume of garbage dumps.

Skills

SecOps Penetration testing, Web Application Security, Cloud Security, Network Security

DevSecOps AWS, Docker, Kubernetes, Snyk, CodeQL, Burpsuite plugin

Technology AWS, Azure, GCP, Docker, Kubernetes, Linux

Techniques Threat modeling, Design Patterns, Source code review, Secure Design, Secure coding

Tools Burpsuite, Wireshark, GNS3, Metasploit, Snort

Database PostgreSQL, MySQL, SQLite3, MongoDB, Neo4j, AWS Aurora, AWS Neptune

Backend Python Django, Python Flask, Golang Gin, SailsJS, NodeJS, ExpressJS, Java Spring Boot

Frontend ReactJS, HTML5, CSS3

Programming Python, Javascript, Java, C/C++, Golang

Extracurricular Activity

Active **Security Blogger**, on Hashnode

Currently **Ranked top 1%**, in TryHackMe Pentesting Lab

Organized **Guest lectures**, with my alma mater on advanced subjects

Was **Project lead**, OSINT platform at Null India

Was **an Active member**, of Intuit's NextGenNetwork

Was **a Representative**, of Intuit's Community Service Group

Honors & Awards

- 2021 **11/100+ rank**, Intuit Redcon CTF 2021
- 2020 **178/5000+ rank**, NahamSecCon CTF 2020
- 2020 **13/600+ rank**, June'gle CTF 2020
- 2020 **4/30+ rank**, Intuit Redcon CTF 2020
- 2019 **NextGenNetwork Champion**, at Intuit's NextGenNetwork
- 2018 **Gold Medalist**, in Computer Science and Engineering, batch 2018 of PES Institute of Technology
- 2014-18 **Professor CNR Rao Merit Scholarhsip award winner**, in undergraduate